

Upsolving past problems

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The point of this document is for training, I will write the problems that I was unable to solve here, then come back later to solve them after having understood the idea of the solution.

CF 1060 Div 2 — C1

Problem statement: <https://codeforces.com/contest/2154/problem/C1>

HMMT Nov 2025 Teams — P5

Kelvin the frog is in the bottom-left cell of a 6×6 grid, and he wants to reach the top-right cell. He can take steps either up one cell or right one cell. However, there is a raccoon in one of the 36 cells uniformly at random, and Kelvin's path must avoid this raccoon. Compute the expected number of distinct paths Kelvin can take to reach the top-right cell.

(If the raccoon is in either the bottom-left or top-right cell, then there are 0 such paths.)

HMMT Nov 2025 Teams — P6

Let P be a point inside triangle $\triangle ABC$ such that $BP = PC$ and $\angle ABP + \angle ACP = 90^\circ$. Given that $AB = 12$, $AC = 16$, and $AP = 11$, compute the area of the concave quadrilateral $ABPC$.

HMMT Nov 2025 Teams — P9

Let a, b and c be positive real numbers such that

$$\begin{aligned}\sqrt{a} + \sqrt{b} + \sqrt{c} &= 7 \\ \sqrt{a+1} + \sqrt{b+1} + \sqrt{c+1} &= 8 \\ (\sqrt{a+1} + \sqrt{a})(\sqrt{b+1} + \sqrt{b})(\sqrt{c+1} + \sqrt{c}) &= 60\end{aligned}$$

Compute $a + b + c$.

HMMT Nov 2025 Guts — P7

Point X lies on diagonal AC of rectangle $ABCD$ such that $AX = 11$, $CX = 1$, and triangle BXD has area 18. Given that $BX < DX$, compute BX .

HMMT Nov 2024 Teams — P6

There are 5 people who start with 1, 2, 3, 4, and 5 cookies, respectively. Every minute, two different people are chosen uniformly at random. If they have a and b cookies and $a \neq b$, the person with more cookies eats $|a - b|$ of their own cookies. If $a = b$, the minute still passes with nothing happening.

Compute the expected number of minutes until all 5 people have an equal number of cookies.

BMO 1 2018 P4/6

Let Γ be a semicircle with diameter AB . The point C lies on the diameter AB and points E and D lie on the arc BA , with E between B and D . Let the tangents to Γ at D and E meet at F . Suppose that $\angle ACD = \angle ECB$.

Prove that $\angle EFD = \angle ACD + \angle ECB$.